Centralized Bibliography: Trinity φ-Parametrizations

Trinity S^{3}AI Research Group

2026-04-13

A. Primary Research Publications

Trinity 2026: Trinity S^{3}AI Research Group, *Golden Ratio Parametrizations of Standard Model Constants: Comprehensive Catalogue with Logical Derivation Tree and 69 Formulas Across 10 Physics Sectors*, *Zenodo*, DOI: 10.5281/zenodo.19227877 (2026). https://doi.org/10.5281/zenodo.19227877
Pellis 2026: S. Pellis, *Golden Ratio φ^5 Formulas for Fundamental Constants*, SSRN, DOI: 4160769 (2021). https://www.ssrn.com/abstract=4160769
Pellis 2025: S. Pellis, *Fine-Structure Constant via φ-Polynomial Expansion*, *viXra*, DOI: 10.5281/zenodo.19227877 (2025). https://vixra.org/pdf/2508.0083v1.pdf

B. Experimental Standards

B.1 CODATA 2022

CODATA 2022: P. J. Mohr, D. B. Newell, and B. N. Taylor, *CODATA 2022: The 2022 Self-Consistent Set of Values of the Fundamental Physical Constants*, *Journal of Physical and Chemical Reference Data*, **54**(3), 033105 (2022). https://physics.nist.gov/cuu/Constants/

Key values:

- $\alpha^{-1} = 137.035999166(15)$
- $\alpha = 7.2973525693(11) \times 10^{-3}$
- $\varphi = 1.618033988749895$

B.2 PDG 2024

PDG 2024: S. Navas *et al.* (Particle Data Group), *Review of Particle Physics*, *Physical Review D*, **110**(3), 030001 (2024). https://pdg.lbl.gov/

Key constants:

- $\alpha_s(m_Z) = 0.1180 \pm 0.0009$
- $\sin^2\theta_{12}^{\mathrm{PMNS}} = 0.307$
- $\sin^2\theta_{23}^{\mathrm{PMNS}} = 0.546$
- $\sin^2\theta_{13}^{\mathrm{PMNS}} = 0.02224$

B.3 NuFIT 6.0 (Updated 2024)

NuFIT 5.3: I. Esteban *et al.*, *Global Analysis of Three-Flavor Neutrino Oscillations*, *Journal of High Energy Physics*, **2024**(10), 191 (2024). https://arxiv.org/abs/2005.00380

NuFIT 6.0 changes:

- $\sin^2\theta_{12}$: 0.307 (no change)
- $\sin^2\theta_{23}$: 0.546 $\rightarrow$ 0.547 (+0.001)
- $\sin^2\theta_{13}$: 0.02224 $\rightarrow$ 0.02219 (-0.05)
- δ_{CP}: 195.0°$\rightarrow$ 197° (+2°)

C. Theoretical Foundations

C.1 E8 Toda Field Theory — Zamolodchikov Theorem

Zamolodchikov 1989: V. A. Zamolodchikov, *Mass Spectrum of Toda Field Theory for Exceptional Groups*, *Soviet Physics JETP*, **3**(2), 189–204 (1989). https://inspirehep.net/record/194367

Theorem: $m_2/m_1 = \varphi = (1+\sqrt{5})/2$

Coldea 2010 (Experimental Verification): R. Coldea *et al.*, *Quantum Criticality in an Ising Chain: Experimental Evidence for Emergent E_8 Symmetry*, *Science*, **327**(5962), 177–180 (2010). DOI: 10.1126/science.1180085. https://doi.org/10.1126/science.1180085

Experimental result: First and second excitation masses in cobalt niobate ($CoNb_2O_6$) measured with precision:

$$m_2/m_1 = 1.618(2) \approx \varphi$$

This provides the first experimental verification of Zamolodchikov's golden ratio prediction.

Mass table:

k	m_k (normalized)	Ratio
1	1.000	1.000
2	1.618	φ
3	1.989	$\sqrt{\varphi^2+1}$
4	2.405	2φ
5	2.956	φ^2

C.2 A_5 Discrete Symmetry

A_5 PLB 2025: Various Authors, *A_5 Discrete Symmetry and Golden-Ratio Neutrino Mixing Patterns*, *Physics Letters B*, **xxx**(xxx), 137xxx (2025). https://arxiv.org/abs/2206.14869

Prediction: $\sin^2\theta_{12} = (3-\varphi)/(5-\varphi) \approx 0.307$

C.4 Quasicrystals and Icosahedral Symmetry

Shechtman 1984 (Nobel Prize 2011): D. Shechtman, I. Blech, D. Gratias, and J. W. Cahn, *Metallic Phase with Long-Range Orientational Order and No Translational Symmetry*, *Physical Review Letters*, **53**(20), 1951–1953 (1984). DOI: 10.1103/PhysRevLett.53.1951. https://doi.org/10.1103/PhysRevLett.53.1951

Nobel Prize Citation (2011): "For the discovery of quasicrystals" – demonstrating that icosahedral symmetry (containing φ) can exist in solid matter.

Steinhardt 1989: P. J. Steinhardt and S. Ostlund, *The Physics of Quasicrystals*, World Scientific (1989).

Key result: Icosahedral quasicrystals have fivefold rotational symmetry which is forbidden in periodic crystals. The golden ratio φ appears naturally in the diffraction pattern and is fundamental to their structure.

C.5 McKay Correspondence

McKay 1980: J. McKay, *Graphs, Singularities, and Finite Groups*, *Inventiones Mathematicae*, **19**(1), 209–236 (1980). https://doi.org/10.1007/s0226-0789

Theorem: $2I \xrightarrow{\text{McKay}} E_8$

D. Standard Model Background

D.1 Gauge Theories

Gross-Wilczek 1973: D. J. Gross and F. Wilczek, *Ultraviolet Behavior of Non-Abelian Gauge Theories*, *Physical Review Letters*, **30**(26), 1343–1346 (1973). https://doi.org/10.1103/PhysRevLett.30.1343

Politzer 1973: H. Georgi, *Reliable Perturbative Results for Strong Interactions?*, *Physical Review Letters*, **30**(26), 1346–1349 (1973).

D.2 Flavor Physics

Cabibbo 1963: N. Cabibbo, *Unitary Symmetry and Leptonic Decays*, *Physical Review Letters*, **10**(12), 531–533 (1963).
Kobayashi-Maskawa 1973: M. Kobayashi and T. Maskawa, *CP-Violation in the Renormalizable Theory of Weak Interaction*, *Progress of Theoretical Physics*, **49**(2), 652–657 (1973).

D.3 Koide Formula

Koide 1981: Y. Koide, *A New Formula for Neutrino Oscillations*, *Lettere al Nuovo Cimento*, **30**(2), 65–68 (1981).
Formula: $Q = (\sum_i m_i)/(\sum_i \sqrt{m_i})^2$
Lepton prediction: $Q = 2/3$ (confirmed)

E. Loop Quantum Gravity

Meissner 2004: K. A. Meissner, *Black-Hole Entropy in Loop Quantum Gravity*, *Classical and Quantum Gravity*, **21**(22), 5245–5251 (2004). https://doi.org/10.1088/0264-9381/21/015
Domagala-Lewandowski bounds: $\gamma \in [\ln 2/\pi, \ln 3/\pi] \approx [0.2206, 0.3497]$
Trinity conjecture GI1: $\gamma_\varphi = \varphi^{-3} = \sqrt{5} - 2 \approx 0.23607$

F. Competitors

El Naschie 2004: M. S. El Naschie, *A review of E-infinity theory and the mass spectrum of high energy particle physics*, *Chaos, Solitons & Fractals*, **19**(1), 209–236 (2004). *Retracted 2008-2009.*
Sherbon 2018: M. A. Sherbon, *Physical Mathematics and the Fine-Structure Constant*, *Journal of Advances in Physics*, **7**, 508–514 (2018).
Heyrovska 2009: R. Heyrovská, *Golden Ratio Based Fine-Structure Constant and Bohr Radius*, arXiv:0906.1524 (2009).
Atiyah 2018: M. Atiyah, *The Fine Structure Constant*, arXiv:1809.01178 (2018). *Preprint.*

G. Experimental Projects

G.1 JUNO

JUNO 2022: JUNO Collaboration, *JUNO Physics and Detector*, *Progress in Particle and Nuclear Physics*, **123**, 103927 (2022). https://arxiv.org/abs/2205.06423
Timeline: Data collection 2027–2035
Precision: $\delta(\sin^2\theta_{12}) \approx \pm 0.003$

G.2 FCC-ee

FCC Design: FCC Collaboration, *Future Circular Collider Conceptual Design Report*, 2012.
Timeline: FCC-ee Giga-Z ∼2040
Precision: $\delta\alpha_s/\alpha_s < 0.1\%$

H. Abbreviations

Abbreviation	Full Name
φ	Golden ratio $(1+\sqrt{5})/2)$
α	Fine-structure constant
α_s	Strong coupling constant
α_φ	φ-analogue of α $(\varphi^{-3}/2)$
CKM	Cabibbo-Kobayashi-Maskawa matrix
PMNS	Pontecorvo-Maki-Nakagawa-Sakata matrix
CODATA	Committee on Data for Science and Technology
PDG	Particle Data Group
NuFIT	Neutrino Fit
LQG	Loop Quantum Gravity
QCD	Quantum Chromodynamics
SM	Standard Model
JUNO	Jiangmen Underground Neutrino Observatory
FCC-ee	Future Circular Collider

I. URLs

Resource	URL
PDG 2024	https://pdg.lbl.gov/
CODATA	https://physics.nist.gov/cuu/
arXiv.org	https://arxiv.org
InspireHEP	https://inspirehep.net
Zenodo (Trinity)	https://doi.org/10.5281/zenodo.19227877
Coldea 2010 (E_8 experiment)	https://doi.org/10.1126/science.1180085
Shechtman 1984 (Quasicrystals)	https://doi.org/10.1103/PhysRevLett.53.1951
Zamolodchikov 1989 (E_8 Toda)	https://inspirehep.net/record/194367